Independent Verification of Public Road Infrastructure Construction Using Multi-Temporal Satellite Imagery

Joshua Castillo and Eric Gutiérrez, Barcelona School of Economics

Barcelona School of Economics

Introduction and Motivation

Public infrastructure spending is a primary driver of economic development, yet ensuring accountability remains a global challenge. While governments maintain procurement databases and transparency portals, independent physical verification of project completion is rare. Traditional field inspections, especially for geographically dispersed rural road networks in countries like the Philippines, are prohibitively expensive, slow, and vulnerable to reporting delays or misaligned incentives. Consequently, there is often a critical gap between a project's reported bureaucratic status and its actual physical progress.

Recent advancements in remote sensing and machine learning offer a scalable solution for independent oversight. Previous research has successfully utilized satellite imagery to predict broad infrastructure quality (Oshri et al., 2018) and extract road networks. Furthermore, modern computer vision techniques, such as Siamese Convolutional Neural Networks and temporal transformers (Zheng et al., 2025), have revolutionized bitemporal change detection. By integrating public procurement metadata with medium-resolution, high-revisit satellite data (Sentinel-2), we can bridge the gap between procurement analytics and remote sensing to create a powerful accountability tool.

This project proposes an end-to-end machine learning system to independently verify the physical progress of government-funded road projects in the Philippines. Rather than predicting simple completion probabilities, we develop independent construction trajectory classifications for verification. By comparing a feature-engineered baseline against a deep learning architecture, we track pixel-level changes across project milestones to evaluate whether the observable satellite evidence aligns with the officially reported status. Ultimately, this system serves as an automated early-warning mechanism, flagging highly anomalous projects to prioritize them for human review by auditors and civil society organizations.

Data

Our analysis integrates three primary data sources to independently verify infrastructure delivery across 2,300 projects. First, administrative records from the Philippine Department of Public Works and Highways (DPWH) provide the baseline procurement data, including project coordinates, construction timelines, and reported completion statuses. Second, we utilize multi-temporal Sentinel-2 satellite imagery (10m resolution) to visually capture the physical state of each project site at distinct intervals throughout its lifecycle. Finally, to isolate relevant structural earthwork from surrounding environmental noise apply a national road network shapefile to generate precise geographic bounding masks. This ensures that our change-detection models strictly evaluate pixel variations within the targeted road corridors.

Methodology

	INPUT	PROCESSING	OUTPUT
1 PROCUREMENT DATA EXTRACTION *Major Road Constructions*	DPWH Transparency Portal API (248K total contracts, 2016–2026)	Filter for major road construction (1+ year contract) with valid coordinates and clear government status (Completed/Terminated).	Filtered Dataset (n = 2.3K projects) with contractId, title, lat/long, address and status
2 SATELLITE & ROAD DATA MATCHING *Construction Milestones*	Sentinel-2 Imagery & OpenStreetMap (OSM) Roads	Query 6 temporal milestones (Pre-Start, Q1, Q2, Q3, Q4, Post-End). Crop images using a spatial buffer and mask pixels outside the OSM road boundary.	6×Images + Road Mask per project for all 2.3K projects.
3 CHANGE DETECTION (PIXEL) *Inter-Milestone Probabilities*	Paired sequential milestone images (t0 -> t1)	Compute pixel-level change probability. Baseline: none, proposed: pre-trained vision transformer (ChangeStar).	5 chronological sets of pixel-level change probabilities per project
4 CHANGE DETECTION (IMAGE/ROAD) *Inter-Milestone Differences*	5 sets of pixel-level change probabilities	Aggregate pixel changes across the masked road ROI. Baseline: normalized absolute pixel difference, ChangeStar: absolute difference filtered strictly for high-confidence pixels (p>0.5).	5 discrete Δ values per project (Baseline vs. ViT)
5 CHANGE DETECTION (PROJECT) *Project Milestone Completion Scores*	5 discrete Δ values per project	Calculate cumulative sum of Δ values, normalized relative to post-completion total change to handle model noise.	5 temporal project completion scores from 0.00 to 1.00 (Baseline vs. ViT)
6 COMPLETION TRAJECTORY TAXONOMY *Project Trajectory Labeling*	5 project completion scores per project	Map scores to infrastructure states, then classify the timelines: Never Started *(max(St) < 0.15)*, Fast Completion *(S25 > 0.70)*, Stalled *(S50 > 0.30 and S100 - S50 < 0.15)*, Completed Weirdly *(S75 < 0.30 and S100 > 0.80)*, Normal Completion	5-category Trajectory Label assigned per project
7 CONSISTENCY ANALYSIS *Independent Verification with Reports*	5-category Trajectory Label vs. 3-category Government Status	Evaluate alignment via a cross-tabulation matrix with Government Status (Completed, Ongoing, Terminated)	Final Project Alignment Classifications (Consistent, Ahead, Review, Potential Discrepancy)
8 REPORT & FLAGGING *Automatic Tagging & Summarization*	Final Project Alignment Classifications	Isolate, count, and extract high-risk projects. Generate automated spatial/temporal plots of anomalous cohorts.	Independent database with specific contracts flagged with a Potential Discrepancy for governance review and reporting on DPWH portal.

Results

Across both models, 98% of the government reported status for major road constructions are consistent. The results from Table 1 show that around 2% of the over 2,300 identified projects have potential discrepancies. An interesting point is that whether a baseline or ViT model was used for probability scoring of pixel changes, the results remain very similar. This suggests the robustness of simple pixel change normalization within road bounds across time. Only 1 project was additionally identified as consistent for the baseline model in further analysis. The only notable difference in favor of ChangeStar was that the mean confidence across constructions curves for change across milestones were higher (prob > 0.80) than those of the baseline (prob < 0.20). Still, the proposed pipeline achieves the same results regardless.

	Trajectory Distribution	Consistency Assessment
Baseline	Normal Completion: 98.5% Never Started: 1.4% Stalled: 0.1%	Consistent: 97.6% **Potential Discrepancy: 2.4%**
ViT (ChangeStar)	Normal Completion: 98.5% Never Started: 1.4% Stalled: 0.0%	Consistent: 97.7% **Potential Discrepancy: 2.3%**

Table 1. Model Comparison For Trajectory Classification and Consistency Assessment

Breaking down the predictions versus reported status, we see most of the consistent results come from completed projects that were normally completed. For those with potential discrepancies, these were projects that showed seemingly the complete opposite: 31 reported as completed but then reportedly never started in the images, and 25 reportedly terminated projects that seemed to reach completion. Table 2 shows the breakdown of these results. These are consistent across both baseline and ViT models.

	Never Started	Stalled	Fast Completion	Completed Weirdly	Normal Completion
Terminated	3	0	0	0	25
Ongoing	0	0	0	0	0
Completed	31	1	0	0	2,317
TOTAL	34	1	0	0	2,340

Table 2. Cross-Tabulation of Reported (row) versus Predicted Status (column) and Consistency
Results: Consistent (green), Review/Ahead (orange) , Potential Discrepancy (red)

Some investigations reveal satellite image quality as a source of discrepancy, leading to dubious predictions from the model. Others show how progress (delta) stalls over time as seen in Figure 1, showing a project in Las Pinas city being reported as completed but being classified as stalled. These are important considerations for the pipeline which can at least flag projects which cannot be independently verified when done remotely via satellite. The rest remain consistent, with not many being considered for review.

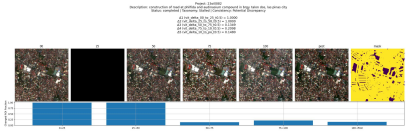


Figure 1. Visualization of Pipeline Output Showing 6 Milestones, Delta Change, Status, Taxonomy, and Consistency Reports

Conclusions

By integrating public procurement data from the Philippine Government with medium-resolution satellite data from Sentinel-2, we were able to create an end-to-end machine learning system that could independently verify the status of government reported road construction projects. This system uses change detection for high-dimensional satellite data across 6 distinct temporal project milestones to end up with construction trajectories to verify consistency with government reports. Results from both simple pixel change computations versus a high performing ViT model show that 98% of the projects in the sample of road construction projects are consistent. Potential discrepancies reveal possible problems in satellite image quality, road masks, or the independent observation of change relative to the status report date. This is still useful since it allows citizens and officials from DPWH to further verify progress on these marked projects. This can be used by DPWH themselves for better transparency on their processes for further human review, either integrating explicitly in the portal or through citizen-led development.

References

Oshri, R., Hu, A., Adelson, P., Chen, X., Dupas, P., Weinstein, J., Burke, M., Lobell, D., & Ermon, S. (2018). Infrastructure quality assessment in Africa using satellite imagery and deep learning. In Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (pp. 616-625).

Zheng, Z., Ermon, S., Kim, D., Zhang, L., & Zhong, Y. (2025). Changen2: Multi-Temporal Remote Sensing Generative Change Foundation Model. IEEE Transactions on Pattern Analysis and Machine Intelligence.